Categories, Proofs, and Processes: Assignment 2

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Question 1

Proof.

$$\begin{aligned}(g_1 \times g_2) \circ (f_1 \times f_2) &= (g_1 \times g_2) \circ \langle f_1 \circ \pi_1, f_2 \circ \pi_2 \rangle \\ &= \langle g_1 \circ f_1 \circ \pi_1, g_2 \circ f_2 \circ \pi_2 \rangle \\ &= (g_1 \circ f_1) \times (g_2 \circ f_2)\end{aligned} \tag{1}$$

$$\begin{aligned}\mathrm{id}_{A_1} \times \mathrm{id}_{A_2} &= \langle \mathrm{id}_{A_1} \circ \pi_1, \mathrm{id}_{A_2} \circ \pi_2 \rangle \\ &= \langle \pi_1 \circ \mathrm{id}_{A_1 \times A_2}, \pi_2 \circ \mathrm{id}_{A_1 \times A_2} \rangle \\ &= \mathrm{id}_{A_1 \times A_2}\end{aligned} \tag{2}$$

□

Question 2

Consider the category $\mathcal{C}$ of 3 objects $A, B, A \times B$ where $\mathcal{C}(A \times B, A) = \{\pi_1\}, \mathcal{C}(A \times B, B) = \{\pi_2\}$, and there are no other arrows other than the identities.

Clearly, $A \times B$ is the initial object.

It has products since the 3 object pairs $(A, B), (A, A \times B), (B, A \times B)$ have the respective products $(A \times B, \pi_1, \pi_2), (A \times B, \pi_1, \mathrm{id}_{A \times B}), (A \times B, \mathrm{id}_{A \times B}, \pi_2)$ in the category.

However, it has no terminal objects, since

$$\mathcal{C}(A, A \times B) = \mathcal{C}(B, A \times B) = \mathcal{C}(A, B) = \mathcal{C}(B, A) = \emptyset$$

so each object has another object it has no arrow from.

It also has no coproducts since A, B have no coproduct in $\mathcal{C}$, since

$$\forall C \in \mathrm{Ob}(\mathcal{C}) : \mathcal{C}(A, C) = \emptyset \vee \mathcal{C}(B, C) = \emptyset$$

so there is no common object for both A, B to have an arrow towards.

Optional

If $P = NP$, every object is an initial and terminal object, product, and coproduct, since there is a reduction between every problem in NP. If $P \neq NP$, P problems are initial objects and products, and NP-complete problems are terminal objects and coproducts.

Question 3

(a)

The initial object is the empty category since there is a unique functor from itself to all other categories. The terminal object is any category with one object (and one identity arrow) since there is a unique functor from every other category to this category.

(b)

The projection functors can be defined as

$$\pi_{\mathcal{C}} :: (A, B) \mapsto A, \quad \pi_{\mathcal{D}} :: (A, B) \mapsto B$$

$$\pi_{\mathcal{C}} :: (f, g) \mapsto f, \quad \pi_{\mathcal{D}} :: (f, g) \mapsto g$$

Proposition 3.1. *For any category $\mathcal{E}$ in* ***Cat*** *with functors $F : \mathcal{E} \to \mathcal{C}, G : \mathcal{E} \to \mathcal{D}$, there is a unique functor*

$$\langle F, G \rangle : \mathcal{E} \to \mathcal{C} \times \mathcal{D}$$

Proof. Given an arbitrary category $\mathcal{E}$ in **Cat** with some object $C \in \mathrm{Ob}(\mathcal{E})$, and some arrow $h \in \mathrm{Ar}(\mathcal{E})$, as well as functors $F : \mathcal{E} \to \mathcal{C}, G : \mathcal{E} \to \mathcal{D}$,

$$F :: C \mapsto F\ C, \quad h \mapsto F\ h$$

$$G :: C \mapsto G\ C, \quad h \mapsto G\ h$$

We can define the functor $\langle F, G \rangle : \mathcal{E} \to \mathcal{C} \times \mathcal{D}$ so

$$\langle F, G \rangle :: C \mapsto (F\ C, G\ C), \quad h \mapsto (F\ h, G\ h)$$

To show its uniqueness, suppose we have some functor $H : \mathcal{E} \to \mathcal{C} \times \mathcal{D}$. By composition, we have that

$$\pi_{\mathcal{C}} \circ H = F$$

$$\pi_{\mathcal{D}} \circ H = G$$

So for any object $C' \in \mathrm{Ob}(\mathcal{E})$, and any arrow $h' \in \mathrm{Ar}(\mathcal{E})$,

$$\pi_{\mathcal{C}} \circ H\ C' = F\ C'$$

$$\pi_{\mathcal{D}} \circ H\ C' = G\ C'$$

$$\pi_{\mathcal{C}} \circ H\ h' = F\ C'$$

$$\pi_{\mathcal{D}} \circ H\ h' = G\ C'$$

which imply that

$$H\ C' = (F\ C', G\ C')$$

$$H\ h' = (F\ h', G\ h')$$

Thus, we have that the functor $H = \langle F, G \rangle$ is unique. $\square$

(c)

Proposition 3.2. *The product functor $- \times - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ satisfies functoriality.*

Proof. Suppose we let F be the product functor. Satisfying *functoriality* means that for any pairs of arrows $(f, g), (f', g')$ where domains and codomains are defined such that $f \circ f', g \circ g'$ are meaningful, and objects A, B in $\mathcal{C}$,

$$F(f, g) \circ F(f', g') = F(f \circ f', g \circ g')$$

$$F(\mathrm{id}_A, \mathrm{id}_B) = \mathrm{id}_{F(A,B)}$$

From Exercise 1, we have

$$(f \times g) \circ (f' \times g') = (f \circ f') \times (g \circ g')$$

$$\mathrm{id}_A \times \mathrm{id}_B = \mathrm{id}_{A \times B}$$

which are equivalent to the *functoriality* axioms, so the product functor satisfies *functoriality*. $\square$

Question 4

(a)

Corollary 4.3. *The singleton $\{*\}$ is not a terminal object in **Rel**.*

Proof. Since any relation $R \subseteq A \times B$ of non-empty sets A, B has size $|R| > 1$ if $B = \{*\} \neq \emptyset$, the arrows will not be unique.

For example, for the singleton $A = \{a\}$, there can be two relations $S, R : A \to \{*\}$,

$$S = \{(a, *)\} \neq \emptyset = R$$

So there is no unique arrow from object A to $\{*\}$.

The singleton $\{*\}$ cannot be a terminal object in **Rel**. $\square$

(b)

The projections $\pi_X : X \sqcup Y \to X, \pi_Y : X \sqcup Y \to Y$ are

$$\pi_X = \{((a,1),a) | (a,1) \in X \sqcup Y\} \cong \{a | (a,1) \in X \sqcup Y\}$$

$$\pi_Y = \{((b,2),b) | (b,2) \in X \sqcup Y\} \cong \{b | (b,2) \in X \sqcup Y\}$$

Proposition 4.4. *The disjoint union is a product in **Rel**.*

Proof. Given an arbitrary set Z with relations $R : Z \to X, S : Z \to Y$, we can define the relation $\langle R, S \rangle : Z \to X \sqcup Y$

$$\langle R, S \rangle = \{(a,(b,1)) | (a,b) \in R\} \cup \{(c,(d,2)) | (c,d) \in S\}$$

Suppose we have some relation $Q : Z \to X \sqcup Y$. By composition, we have

$$\pi_X \circ Q = R$$

$$\pi_Y \circ Q = S$$

Then we have

$$\forall (a,b) \in R, \exists (b,1) \in X \sqcup Y : (a,(b,1)) \in Q \wedge b \in \pi_X$$

$$\forall (a',b') \in S, \exists (b',2) \in X \sqcup Y : (a',(b',2)) \in Q \wedge b' \in \pi_Y$$

$$\forall (b,n) \in X \sqcup Y, \exists a \in Z : (a,(b,n)) \in Q \implies (n = 1 \wedge (a,b) \in R) \vee (n = 2 \wedge (a,b) \in S)$$

Respectively, we can then infer

$$Q \supseteq \{(a,(b,1)) | (a,b) \in R\}$$

$$Q \supseteq \{(a',(b',2)) | (a',b') \in S\}$$

$$Q \subseteq \{(a,(b,1)) | (a,b) \in R\} \cup \{(a',(b',2)) | (a',b') \in S\}$$

Thus, $Q = \langle R, S \rangle$ is the unique relation, and the disjoint union is a product in **Rel**. $\square$

(c)

Proposition 4.5.

$$\boldsymbol{Rel} \cong \boldsymbol{Rel}^{op}$$

Proof. For all sets A, B and all relations $\{(a_i, b_i)\}_i \subseteq A \times B$ with $a_i \in A, b_i \in B$ in **Rel**, define the functors such that

$$F :: A \mapsto A, \quad \{(a_i, b_i)\}_i \mapsto \{(b_i, a_i)\}_i$$
$$G :: A \mapsto A, \quad \{(b_i, a_i)\}_i \mapsto \{(a_i, b_i)\}_i$$

Then we have

$$G \circ F\ \{(a_i, b_i)\}_i = G\ \{(b_i, a_i)\}_i = \{(a_i, b_i)\}_i = \mathrm{id}_{\mathbf{Rel}}\ \{(a_i, b_i)\}_i$$
$$F \circ G\ \{(b_i, a_i)\}_i = F\ \{(a_i, b_i)\}_i = \{(b_i, a_i)\}_i = \mathrm{id}_{\mathbf{Rel}^{\mathrm{op}}}\ \{(b_i, a_i)\}_i$$

Then we have $G \circ F = \mathrm{id}_{\mathbf{Rel}}, F \circ G = \mathrm{id}_{\mathbf{Rel}^{\mathrm{op}}}$ as desired. □

(d)

Corollary 4.6. *The coproduct in **Rel** is the disjoint union.*

Proof. Since the product in **Rel** is the disjoint union, and $\mathbf{Rel} \cong \mathbf{Rel}^{\mathrm{op}}$, the product in $\mathbf{Rel}^{\mathrm{op}}$ is also the disjoint union. Thus, the coproduct in **Rel** is also the disjoint union. □

Question 5

Proposition 5.7. *The functor $F : \boldsymbol{Mat}_{\mathbb{R}} \to \boldsymbol{FDVector}_{\mathbb{R}}$ is an equivalence.*

Proof. Define the object mapping as

$$F :: n \mapsto \mathbb{R}^n$$

and the arrow mapping

$$M : m \to n \in \mathbb{R}^{n \times m}$$
$$F\ M : \mathbb{R}^m \to \mathbb{R}^n := (x_1, ..., x_m)^T \mapsto M(x_1, ..., x_m)^T$$

for $M \in \mathbb{R}^{n \times m}$.

The functor is full since for every arrow M, M' in $\mathbf{Mat}_{\mathbb{R}}$,

$$\begin{aligned} F\ M = F\ M' &\implies \forall \vec{x} \in \mathbb{R}^m : M\vec{x} = M'\vec{x} \\ &\implies \forall \vec{x} \in \mathbb{R}^m : (M - M')\vec{x} = \vec{0} \\ &\implies \forall \vec{x} \in \mathbb{R}^m : M - M' = \vec{0} \\ &\implies M = M' \end{aligned} \tag{3}$$

and is faithful since for every arrow in $\mathbf{FDVector}_{\mathbb{R}}$

$$N : \mathbb{R}^m \to \mathbb{R}^n := (x_1, ..., x_m)^T \mapsto M(x_1, ..., x_m)^T$$

there exists the arrow $M \in \mathbb{R}^{n \times m}$ in $\mathbf{Mat}_{\mathbb{R}}$. The functor is essentially surjective since for every object $R^n \in \{\mathbb{R}^i\}_i, i \in \mathbb{N}$ of $\mathbf{FDVector}_{\mathbb{R}}$, there is an object $n \in \mathbb{N}$ of $\mathbf{Mat}_{\mathbb{R}}$ where $F(n) \cong R^n$.

Thus, the functor is an equivalence. □

Question 6

Proposition 6.8. *The functor $F : \textbf{Set} \to \textbf{Mon}$ as defined is faithful but not full.*

Proof. The functor is faithful since for all arrows f, g in **Set**,

$$\begin{aligned} F\ f = F\ g &\implies \forall x_i \in X : F\ f(x_1...x_n) = F\ g(x_1...x_n) \\ &\implies \forall x_i \in X : f(x_1)...f(x_n) = g(x_1)...g(x_n) \\ &\implies \forall x_i \in X : f(x_i) = g(x_i) \\ &\implies f = g \end{aligned} \tag{4}$$

To show that the functor is not full, consider the free monoid on the singleton $\{1\}$, isomorphic to $(\mathbb{N}, +, 0)$. Consider also the function between two singletons $f : \{1\} \to \{1\}$.

Clearly, there is only one possible function between two singletons, since it is a terminal object.

However, there exists multiple monoid homorphisms $h : (\mathbb{N}, +, 0) \to (\mathbb{N}, +, 0)$. Other than one where the underlying function $|h| = \mathrm{id}_{\mathbb{N}}$, we could have $|h|(x) = 2x$. Note that this is a valid monoid homomorphism which preserves the structure.

Thus, there exists monoid homorphisms which cannot be mapped from any function using the functor. The functor is not full.

□